

MANUFACTURING TOOL TECHNOLOGY
SEMESTER -3(Mechanical/Metallurgy)
Week # 01

Safety standards Practiced in the machine Shop:

i. Wear Protective Gear:

Safety glasses, ear protection, gloves, safety helmet, safety shoes, and flame-resistant clothing.

ii. Machine and Electrical Safety:

- Use guards on machines and regularly check and maintain them.
- Turn off and secure machines when fixing them.
- Ground electrical equipment, use circuit breakers, cords and replace damaged.

iii. Keep the Area Safe:

- Keep the workspace clean and organized.
- Ensure good lighting, mark danger zones, and keep walkways and exits clear.

iv. Chemical and Fire Safety:

- Properly label and store chemicals and use ventilation for fumes.
- Keep fire extinguishers nearby and train everyone on how to use them.
- Store flammable materials safely.

v. Training, Procedures, and Emergency Plans:

- Conduct regular safety training and follow standard procedures.
- Report unsafe conditions and perform safety audits.
- Keep a first aid kit, train in first aid and practice evacuation plans.

Importance of housekeeping:

i. Safety:

- Prevents accidents and injuries from tools.
- Reduces fire risks.

ii. Efficiency:

- Makes it easier to find tools and materials.
- Keeps work flowing smoothly and meets regulatory standards.

iii. Machine Care:

- Keeps machines and tools in good condition.
- Prevents damage from dust and debris (flying particles).

iv. Cleanliness:

- Keeps products clean and free from contamination.
- Creates a nicer, more professional workplace for employees and visitors.

v. Health and Environmental Impact:

- Reduces dust and chemicals that can cause health issues.
- Ensures proper disposal and recycling, keeping pollutants contained

Good Shop Floor Practices (5S):

What is 5S?

- 5S is a method for organizing a workspace to make it cleaner, safer, and more efficient.
- Each "S" represents a step:

1. **Sort (Seiri):** Remove unnecessary items from the workspace.
- ii. **Set in Order (Seiton):** Organize tools and materials so they are easy to find and use.
- iii. **Shine (Seiso):** Clean the workspace thoroughly to spot issues early.
- iv. **Standardize (Seiketsu):** Establish rules and procedures for maintaining cleanliness and organization.
- v. **Sustain (Shitsuku):** Make 5S habit and continuously improve.



Importance of 5S on the Shop Floor:

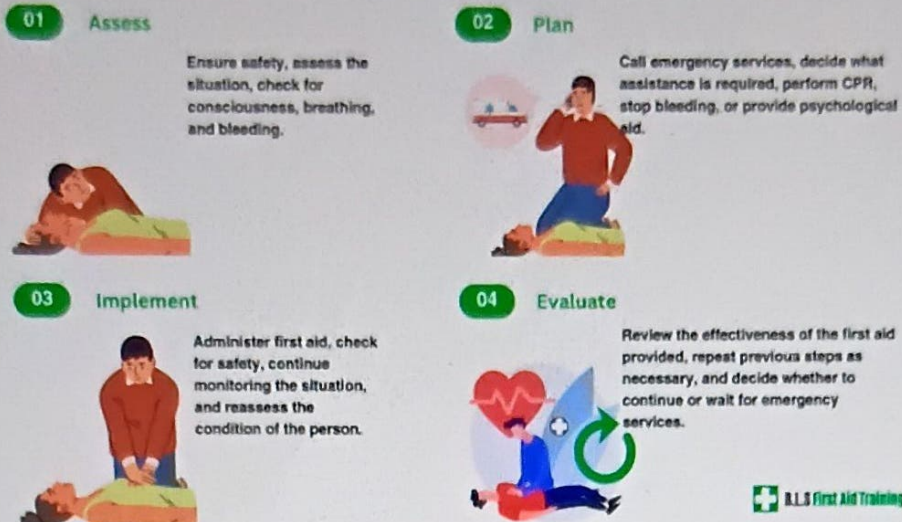
1. **Efficiency:** Reduces time wasted looking for tools and materials.
2. **Safety:** Creates a safer environment by removing hazards or danger.
3. **Quality:** Improves product quality by reducing errors and defects.
4. **Morale:** Boosts employee satisfaction and pride in their work environment.
5. **Cost Savings:** Reduces waste and maintenance costs.

Understand the Principles of First Aid:

- i. **Save Lives:** Priority is to keep the person alive.
- ii. **Avoid Making Things Worse:** Prevent injuries from getting worse.
- iii. **Promote Recovery:** Help the person heal and feel better.
- iv. **Protect:** Keep them safe from further harm.
- v. **Get Help:** Call emergency services if needed.
- vi. **Comfort:** Offer comfort and support.

4 Essential Steps of First Aid:

Assess, Plan, Implement, Evaluate



Preventive measures to be taken during Fire and Electrical emergency :

Fire Emergency:

- **Install Smoke Alarms:** Have working smoke alarms in your home.
- **Keep Fire Extinguishers:** Store fire extinguishers where it is easy to reach.
- **Plan Escape Routes:** Know how to exit quickly in a fire.
- **Cheek Machines:** Regularly inspect machines and appliances.

Electrical Emergency:

- **Turn Off Power:** If safe, turn off the power during an electrical problem.
- **Use Fire Extinguisher:** Use a fire extinguisher made for electrical fires, not water.
- **Call for Help:** Dial emergency services if needed.
- **Avoid Overloading:** Don't overload outlets or cords.

Types of Fire extinguisher (Class A, B, C, D & K):

i. **Class A Fire Extinguisher:** Used to extinguish fires caused due to combustion of cloth, wood, rubber, paper and plastics.

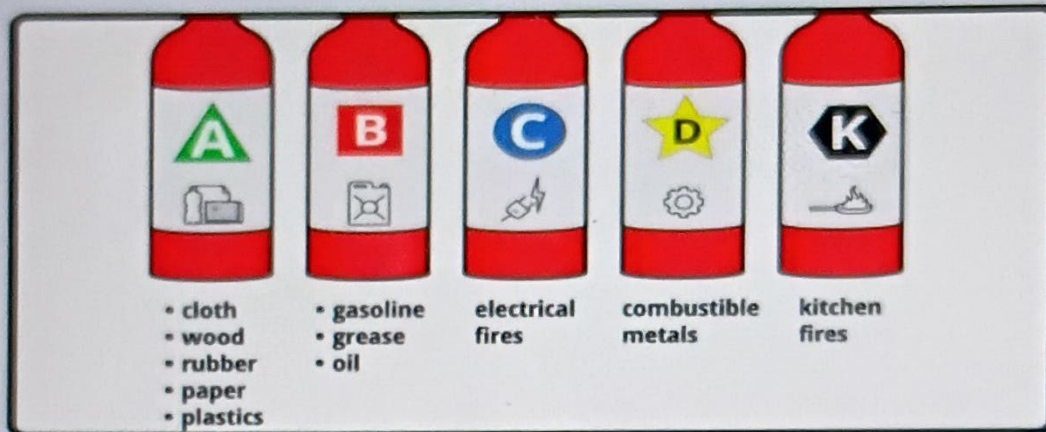
ii. **Class B Fire Extinguisher:** Used to extinguish fires caused due to combustion gasoline, grease and oil.

iii. **Class C Fire Extinguisher:** Used to extinguish fires caused due to electrical equipment and appliances.

iv. **Class D Fire Extinguisher:** Used to extinguish fires caused due to combustion metals like magnesium and sodium.

v. **Class K Fire Extinguisher:** Used to extinguish fires caused due to combustion of cooking oils and fats, typically in kitchens.

Types of Fire Extinguishers



Knowledge about Safety and Environment Regulations:

- i. **Workplace Safety:** Ensures safe working conditions with required safety training and equipment.
- ii. **Environmental Protection:** Controls pollution and protects air and water quality through regulations.
- iii. **Chemical Safety:** Requires clear labelling and information on hazardous chemicals in the workplace..
- iv. **Emergency Planning:** Mandates reporting and planning for handling hazardous substances during emergencies.
- v. **Waste Management:** Regulates the proper disposal and handling of industrial waste to protect the environment.